BRAIN-UMAP: The genetic intersection between neuroscience, neurology, psychiatry, and oncology

Sonali Arora[1], Nicholas Nuechterlein[2], Siobhan S Pattwell[1], Eric Holland[1]
[1]Fred Hutchinson Cancer Research Center, Seattle, WA; [2]University of Washington, Seattle, WA

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Abstract: Incorporating RNA-Sequencing data from gliomas, neurological and psychiatric disorders (such as Parkinson's, Huntington's) and contrasting it with gene expression data for GTEx-defined normal brain regions, can increase our understanding of all brain diseases. We apply uniform manifold approximation and projection (UMAP), a non-linear dimension reduction tool, to bulk gene expression data and build a BRAIN-UMAP and further overlay copy number and mutation calls on the same to understand the landscape of brain diseases.

Fig 1: Constructing the BRAIN-UMAP

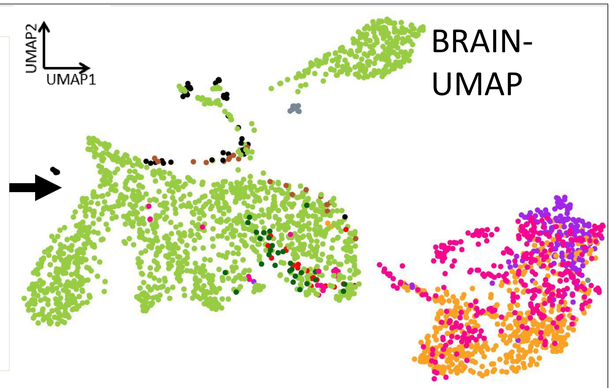
Dataset Accession #	No of Control Samples	No of Disease Samples	Disease Name
SRP033725	30	31	Bipolar disorder
SRP051844	49	20	Huntington's
SRP058181	44	29	Parkinson
ERP001304	9	9	Schizophrenia
SRP035524	11	24	Schizophrenia
SRP032476	0	13	PNET
TCGA-GBM	0	175	Gliomas
TCGA-LGG	0	532	Gliomas
CGGA	0	270	Gliomas
GTEX	1409	0	Normal Brain

Subset for shared genes

Normalized to Transcripts per million (TPM)

Construct BRAIN-UMAP

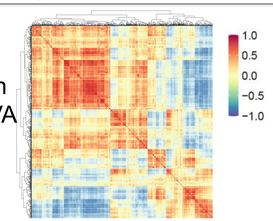
Overlay other types of genomic data



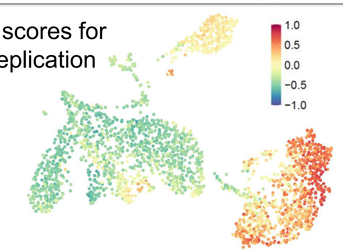
• Bipolar disorder • Huntington's disease • Schizophrenia • TCGA−GBM • CGGA
• PNET • Parkinson's disease • Normal Brain • TCGA−LGG

Fig 2: Different Pathways & Immune cell types regulated in Gliomas

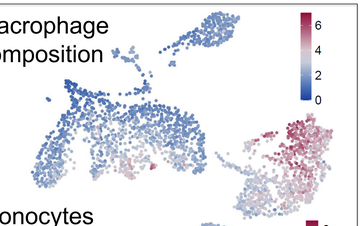
Heatmap showing correlation of all GSVA pathway scores



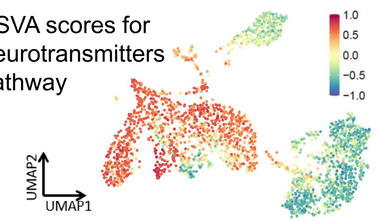
GSVA scores for DNA replication



Macrophage composition



GSVA scores for Neurotransmitters Pathway



Other Pathways up-regulated in Gliomas
- PI3K/AKT signaling (Reactome)
- Notch signaling Pathway (KEGG)
- Signaling by WNT (Reactome)
- RAS Pathway (Biocarta)
- Cell cycle (KEGG)
- Translation (Reactome)

Monocytes composition

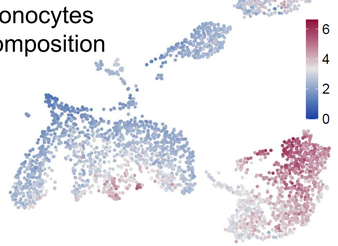


Fig 3: Delineating regions of Healthy Brain

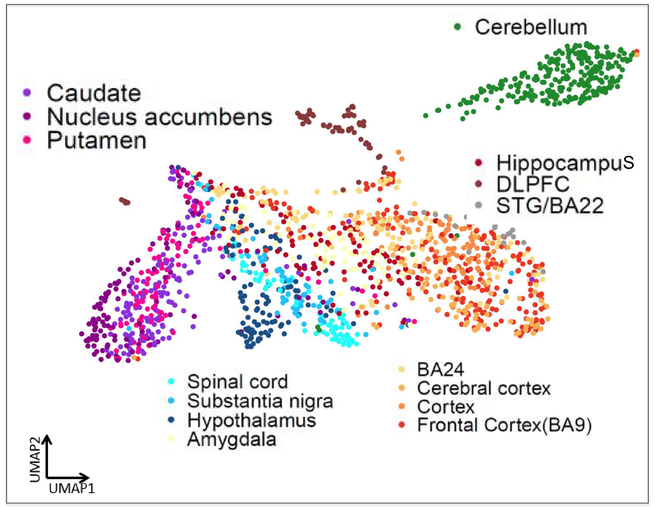


Fig 4: TCGA Glioma landscape

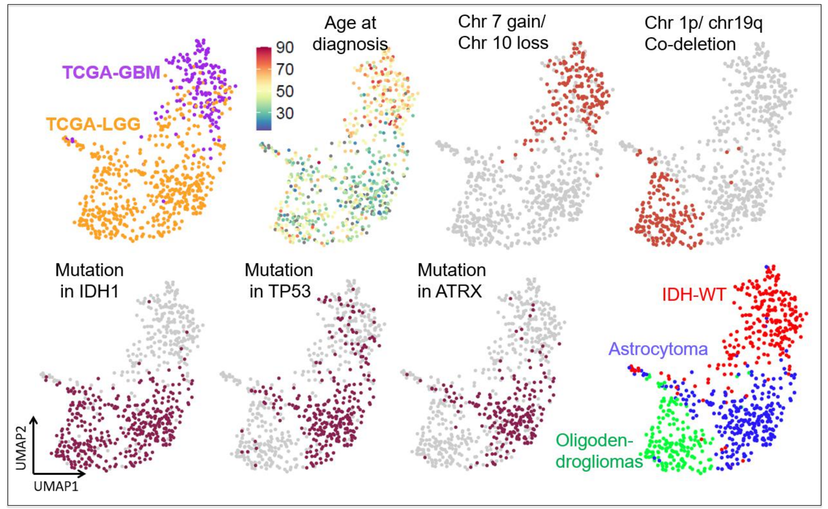
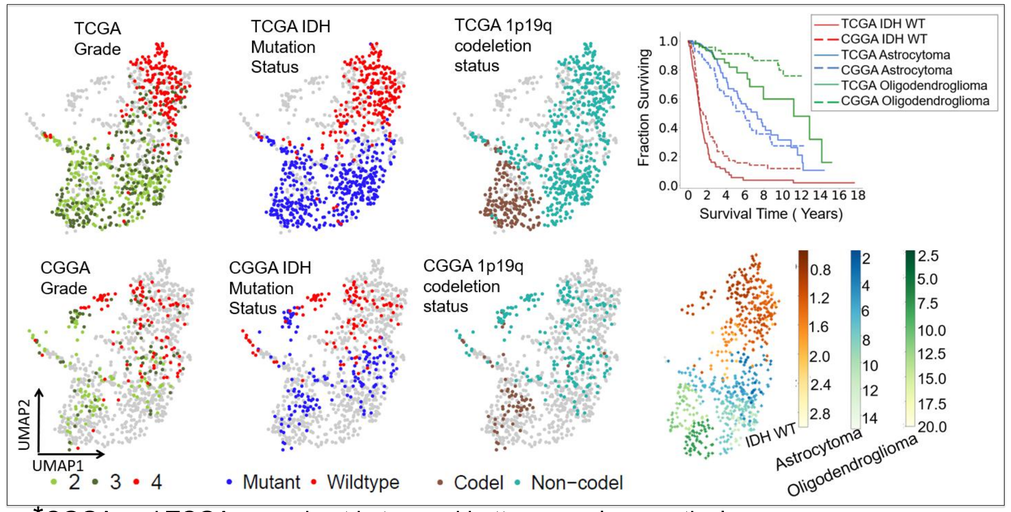


Fig 5: Glioma landscape is consistent across TCGA and CGGA



*CGGA and TCGA greyed out in top and bottom panel respectively

References

1. Collado-Torres L, Nellore A, Kammers K, Ellis SE, Taub MA, Hansen KD, Jaffe AE, Langmead B, Leek JT (2017). "Reproducible RNA-seq analysis using recount2." Nature Biotechnology.
2. Jake R Conway, Alexander Lex, Nils Gehlenborg UpSetR: An R Package for the Visualization of Intersecting Sets and their Properties doi: https://doi.org/10.1093/bioinformatics/btx364
3. Hänzelmann S, Castelo R, Guinney J (2013). "GSVA: gene set variation analysis for microarray and RNA-Seq data." BMC Bioinformatics, 14, 7
4. Grossman, Robert L., Heath, Allison Pet al. (2016) Toward a Shared Vision for Cancer Genomic Data. New England Journal of Vivian J, Rao AA, Nothaft FA, et al. (2017) Toil enables reproducible, open source, big biomedical data analyses. Nature biotechnology.
5. Goldman, M.J., Craft, B., Hastie, M. et al. Visualizing and interpreting cancer genomics data via the Xena platform. Nat Biotechnol (2020). https://doi.org/10.1038/s41587-020-0546-8